

Data Analytics in Cloud Computing

Exercise 3

Snowflake & Superset Lab

Cloud Data Warehouse • Apache Superset • Employee Retention Risk Analysis



Introduction

In this lab you will build the same analytics workflow across a managed cloud data platform and a hosted BI service: load HR data into Snowflake, model an analysis view, connect it to Apache Superset hosted on Preset, and interpret attrition drivers through a dashboard. **Snowflake** is a cloud-native data platform used for storing, processing, and analyzing data at scale. **Apache Superset** is an open-source BI and data visualization platform, while **Preset** is the managed cloud service that hosts and operates Superset for you.

From a cloud service-model perspective, **Preset is SaaS**: you consume the finished analytics application through the browser and the provider manages the software stack. **Snowflake sits between SaaS and PaaS**. It is often described as *DBaaS or DWaaS* because the full database platform is delivered as a service, but it also has strong *PaaS characteristics*: Snowflake manages infrastructure, storage, scaling, upgrades, and much of performance engineering, while you still design schemas, write SQL, manage roles, and operate data workloads. In practice, it is best understood as a managed data platform that is more configurable than typical SaaS but less infrastructure-heavy than classic PaaS.

How to copy code (important)

One-liner commands and short SQL statements in this document are safe to copy and run.

For longer SQL blocks, copy from the provided script file to avoid formatting issues and to keep object names consistent.

Repo / lab files used in this lab:

- snowflake_setup.sql
- employee_history.csv
- pulse_survey.csv
- hr_activity.csv
- office_lookup.csv
- current_workforce.csv

Prerequisites

- A Snowflake trial account with permission to create or use a warehouse, database, and schema.
- A Preset account or trial with access to create datasets, charts, and dashboards.
- The five CSV files included with the lab, plus the SQL setup script.
- Basic SQL skills and familiarity with browser-based analytics tools.

Cloud offerings overview (Snowflake and Preset)

Snowflake platform characteristics:

- Managed storage and compute separation - scale warehouses independently from stored data.
- Cloud data warehousing and analytics - SQL-based analysis, views, roles, and governed data sharing.
- Service model - commonly described as DBaaS or DWaaS, with PaaS-like operational characteristics.
- Operational responsibility split - Snowflake manages infrastructure, patching, elasticity, and many optimization tasks; you manage data models, SQL logic, access control, and cost-aware usage.

Preset / Apache Superset characteristics:

- Apache Superset provides dashboards, charts, filters, datasets, and SQL exploration for BI workloads.
- Preset hosts and manages Superset as a cloud service, removing the need to deploy and operate the BI stack yourself.
- Service model - SaaS, because users consume a ready-to-use application through the browser.
- Operational responsibility split - Preset manages hosting and application operations; you manage datasets, chart logic, dashboard design, and user-facing analytics content.

Lab files

Use the following files for the employee retention scenario.

File	Rows	Purpose
employee_history.csv	650	Historical employees with attrition outcome
pulse_survey.csv	650	Latest survey snapshot per employee
hr_activity.csv	650	Activity and operational measures
office_lookup.csv	10	Office and region lookup
current_workforce.csv	180	Bonus dataset for rule-based risk scoring
snowflake_setup.sql	1 script	Optional DDL plus required view creation SQL

Recommended folder name on your local machine: employee_retention_lab

Dataset and deliverables

The dataset is synthetic and represents employee history, survey, operational, and office-location data for an HR attrition analysis use case.

Expected deliverables:

- One Snowflake analysis view named EMPLOYEE_RETENTION_LAB_VW.
- One Preset dashboard named Employee Retention Risk Dashboard.
- One short written answer to the business questions at the end of the lab.

Task 1 - Snowflake setup and data loading

Learning objectives

- Create or select the warehouse, database, and schema used for the lab.
- Load the five CSV files into Snowflake tables.
- Validate that all source tables contain the expected number of rows.

Step 1: Create the Snowflake environment

In Snowsight, open a new SQL worksheet and run the environment setup section from the SQL file. If your trial already includes a usable warehouse, you can keep using it instead of creating a new one.

```
CREATE OR REPLACE WAREHOUSE HR_LAB_WH
  WAREHOUSE_SIZE = 'XSMALL'
  AUTO_SUSPEND = 60
  AUTO_RESUME = TRUE;

CREATE OR REPLACE DATABASE HR_LAB_DB;
CREATE OR REPLACE SCHEMA HR_LAB_DB.RETENTION_LAB;

USE WAREHOUSE HR_LAB_WH;
USE DATABASE HR_LAB_DB;
USE SCHEMA RETENTION_LAB;
```

Checkpoint: your worksheet context should now point at HR_LAB_DB.RETENTION_LAB.

Step 2: Load the CSV files

Use Snowsight's data loading UI to import each CSV file. Either let Snowflake infer the schema or create empty tables first with the SQL script and then load into those tables.

- Use these table names exactly: EMPLOYEE_HISTORY, PULSE_SURVEY, HR_ACTIVITY, OFFICE_LOOKUP, CURRENT_WORKFORCE.
- Check row counts after each file loads.
- Do not continue until all five tables are present in the RETENTION_LAB schema.

Validation queries:

```
SELECT COUNT(*) AS employee_rows FROM employee_history;
SELECT COUNT(*) AS survey_rows FROM pulse_survey;
SELECT COUNT(*) AS activity_rows FROM hr_activity;
SELECT COUNT(*) AS office_rows FROM office_lookup;
SELECT COUNT(*) AS current_rows FROM current_workforce;
```

Expected row counts: employee_history 650, pulse_survey 650, hr_activity 650, office_lookup 10, current_workforce 180.

Step 3: Create the analysis view

Run the view creation section from the SQL file. The key output is EMPLOYEE_RETENTION_LAB_VW. This is the main dataset used by Preset.

```
CREATE OR REPLACE VIEW employee_retention_lab_vw AS
SELECT
  e.employee_id,
  e.first_name || ' ' || e.last_name AS full_name,
  e.gender,
  e.age,
  e.department,
  e.job_role,
  e.job_level,
  e.business_unit,
  o.region,
  o.office_city,
  ROUND(DATEDIFF('day', e.hire_date, COALESCE(e.exit_date, CURRENT_DATE())) / 365.25, 1) AS
tenure_years,
  e.monthly_income_usd,
  CASE
    WHEN e.monthly_income_usd < 5000 THEN 'Under 5k'
    WHEN e.monthly_income_usd < 8000 THEN '5k-8k'
    WHEN e.monthly_income_usd < 11000 THEN '8k-11k'
    ELSE '11k+'
  END AS income_band,
  e.remote_type,
  e.travel_frequency,
  e.overtime_flag,
  e.performance_rating,
  e.promotion_last_24m,
  e.attrition,
  CASE WHEN e.attrition = 'Yes' THEN 1 ELSE 0 END AS attrition_flag,
  s.engagement_score,
  s.manager_support,
  s.career_growth,
  s.work_life_balance,
  s.stress_score,
```

```

a.training_hours_ytd,
a.absence_days_ytd,
a.projects_supported,
a.recognition_events_ytd
FROM employee_history e
LEFT JOIN pulse_survey s ON e.employee_id = s.employee_id
LEFT JOIN hr_activity a ON e.employee_id = a.employee_id
LEFT JOIN office_lookup o ON e.office_code = o.office_code;

```

Sanity check queries:

```

SELECT COUNT(*) FROM employee_retention_lab_vw;

SELECT
    ROUND(100.0 * SUM(attrition_flag) / COUNT(*), 1) AS attrition_rate_pct
FROM employee_retention_lab_vw;

SELECT department,
    ROUND(100.0 * SUM(attrition_flag) / COUNT(*), 1) AS attrition_rate_pct
FROM employee_retention_lab_vw
GROUP BY 1
ORDER BY 2 DESC;

```

Expected overall attrition rate: about 16.8%.

Task 2 - Connect Preset to Snowflake

Learning objectives

- Create a Preset dataset from the Snowflake analysis view.
- Check that temporal fields are recognized correctly.
- Define the saved metric used across charts.

Step 1: Create the dataset

- In Preset, connect your Snowflake database or select an existing Snowflake connection.
- Fill the data as asked. Under account make sure to enter ACCOUNT IDENTIFIER from snowflake
- Create a dataset from EMPLOYEE_RETENTION_LAB_VW.
- Mark temporal fields correctly if Preset infers them incorrectly.

Step 2: Create the dashboard metric

Create one saved metric named attrition_rate_pct using the expression below.

```
SUM(attrition_flag) / COUNT(*)
```

Task 3 - Build the dashboard in Preset

Learning objectives

- Build KPI tiles, comparison charts, and a heatmap from the Snowflake-backed dataset.
- Add dashboard filters for common HR slicing dimensions.
- Use the dashboard to identify high-risk attrition segments.

Step 1: Create the dashboard

Create a dashboard named Employee Retention Risk Dashboard and save all charts to that dashboard.

Step 2: Build the required charts

Item	Chart type	Build spec	Notes
KPI 1	Big Number	Metric: COUNT(*)	Title: Employees in historical sample
KPI 2	Big Number	Metric: attrition_rate_pct	Format as percentage; Title: Attrition rate
Chart 1	Horizontal Bar	Y-Axis: department Metric: attrition_rate_pct	Sort descending by metric, so highest is on top. Format to percentage
Chart 2	100% Stacked Bar	X-Axis: overtime_flag Metric: COUNT(*) Dimension: attrition	Shows proportion leaving vs staying
Chart 3	Horizontal Bar	Y-Axis: travel_frequency Metric: attrition_rate_pct	Sort descending
Chart 4	Heatmap	X: stress_score Y: work_life_balance Metric: attrition_rate_pct	Sort by both axis ascending.
Chart 5	Table	Dimensions: department, region, overtime_flag, travel_frequency Metrics: COUNT(*), attrition_rate_pct	Sort by attrition_rate_pct descending; limit to 15 rows

Step 3: Add dashboard filters

- department
- region
- remote_type

Recommended layout: place the two KPI tiles at the top, the comparison charts in the middle, the heatmap on the lower left, and the table on the lower right.

Task 4 - Answer the business questions

Use the completed dashboard to answer the following questions.

- Which department has the highest attrition rate?
- How different is attrition for employees with overtime versus no overtime?
- Does travel frequency appear to matter?
- Which stress and work-life balance combinations look most concerning in the heatmap?
- What are the top three actions HR should investigate first?

Expected findings

- Overall attrition is roughly 16% to 17%.
- Customer Support and Operations are the highest-risk departments.
- Employees reporting overtime have materially higher attrition than those who do not.

- Frequent travel is associated with higher attrition than rare travel.
- Low work-life balance combined with high stress creates the hottest risk cells.

Optional extension - Score the current workforce

Run the bonus view in the SQL file to create CURRENT_WORKFORCE_RISK_VW. Then build a simple table in Preset with full_name, department, region, risk_score, and risk_band. Filter to risk_band = 'High' and sort by risk_score descending.

```
SELECT CONCAT(first_name, ' ', last_name) as full_name, department, region, risk_score, risk_band
FROM current_workforce_risk_vw
ORDER BY risk_score DESC
LIMIT 20;
```

Cleanup

- Suspend or stop the warehouse when you finish the lab if your trial does not auto-suspend it quickly enough.
- Keep the database and schema if you plan to extend the dashboard; otherwise remove lab objects to avoid clutter.